

POCKET PRODUCER: SESSION-CONDITIONED RETRIEVAL AS USER-CONTROLLED PROPOSAL

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ABSTRACT

We present Pocket Producer, a prototype that operationalizes *session-conditioned continuation retrieval* inside the Audiotool DAW. A five-signal logistic fusion, trained with BPR on pack co-membership weak labels from the Freesound Loop Dataset, serves as the cold-start candidate generator; a production variant of the same recipe is deployed in the system. Four lines of evidence show this proxy should not decide autonomously: (1) the apparent fusion gain is label-sensitive—+4.0 pp over frozen cosine on hard_similar under mixed uploader/pack labels, but −0.03 pp under pack-only labels; (2) leave-one-signal-out ablation shows regime-dependent contributions—tempo helps on hard_similar but hurts on hard_tempo_key; (3) on a pre-committed held-out split the fusion—cosine advantage on hard_similar swings from +1.5 pp (CI crossing zero, LAION-CLAP) to +9.2 pp (CI above zero, MS-CLAP)—representation-sensitive; (4) an exploratory pilot shows a descriptive divergence between weak-label signal rankings and human consensus. Pocket Producer therefore treats every recommendation as a *proposal*: candidates are previewable, insertion is reversible, and feedback is logged for future personalization (not yet evaluated).

1. INTRODUCTION

When a musician returns to an unfinished loop-based project, the question is not “what sounds similar?” but “what would be useful as the next step?” Loop compatibility [1–4], stem coherence [5–7], session-conditioned generation [8], and reversible co-creation [9] all address parts of this problem. We define **continuation utility** as the property that a candidate helps the musician make progress in the current session; unlike similarity it is asymmetric and context-dependent.

We contribute: (1) a DAW-integrated prototype that deploys the evaluated five-signal fusion as a selectable ranker and treats retrieval as user-controlled proposal, not automatic decision; (2) empirical evidence that weak-label proxy scores are sensitive to label construction, regime-dependent, uncertain on held-out data, and possibly mis-

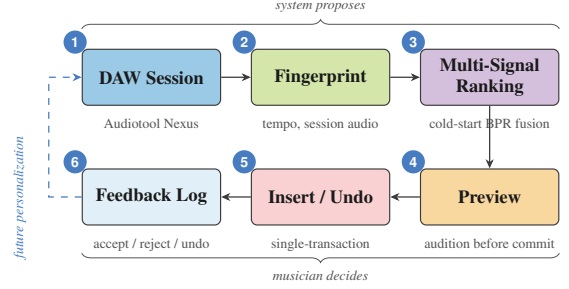


Figure 1. Pocket Producer pipeline: the system proposes (steps 1–3); the musician decides (steps 4–6).

aligned with human judgment—motivating keeping creative authority with the musician.

2. SYSTEM

The live Audiotool session (Figure 1) is read via the Nexus SDK and mapped to a session fingerprint (tempo, and CLAP embeddings [10] of up to eight unique region-referenced samples). A five-signal logistic fusion trained with BPR loss [11] ranks candidates across timbral similarity (CLAP cosine to session regions, mean- and max-pooled), tempo match, key compatibility, and tag overlap. The deployed ranker follows the same recipe but is a separate production model with four stated deltas: MS-CLAP space (offline: LAION-CLAP Music); source-weighted BPR weights on the pack-only train side (hard_similar $\times$ 2.0, mean over 5 seeds); key constant zero online (no key signature in Audiotool documents); tag context from session fragments, not pack siblings. Given the results below, the *default* ranker is the transparent rules baseline; fusion remains selectable and falls back to rules without session audio. Candidates are surfaced for preview—with per-signal evidence—before any session state changes; accepted candidates are inserted in a single Nexus transaction with undo, and rejected or undone events are logged. This responds to Section 4’s findings: the proxy must not substitute its ranking for the musician’s judgment.

3. WEAK-LABEL FUSION

From the FSLD [12] we kept only sounds with a true Freesound pack (1,797 items, 605 packs), discarding items grouped only by uploader. Each eligible pack with at



Table 1. LOSO ablation: accuracy change (pp) when one signal is removed. Pack-only train side (Section 3), mean over 5 splits.

Signal removed	easy	hard_sim.	hard_tk
audio_cos_mean	-0.32	-0.36	-1.04
audio_cos_max	+0.17	-0.30	-0.15
tempo	-0.27	-0.96	+0.68
key	-0.21	+0.94	+1.21
tag_jaccard	+0.09	-0.79	-0.07

least three items yields one leave-one-out query per item, with same-pack siblings as positives and cross-pack items as negatives in three difficulty tiers (*easy*: tempo/key mismatch; *hard_tempo_key*: compatible tempo and key; *hard_similar*: highest cosine to the positive from another pack). A 121-source held-out split (seed 20260810) was committed to version control together with the evaluation protocol *before* any evaluation ran. Development uses only the 484 train sources; the 171 with ≥ 3 items form queries (981 examples), the rest contribute negatives only. LAION-CLAP Music [13] (512-dim) provides the embedding backbone.

On the train side (mean over 5 source-grouped splits), frozen cosine reaches 0.929 overall and 0.814 on *hard_similar*; five-signal fusion reaches 0.925 and 0.813. Under pack-only labels the fusion advantage is gone: -0.03 pp on *hard_similar*, and fusion trails cosine on *hard_tempo_key* by 2.1 pp on average (negative in all 5 splits). These development-set numbers are descriptive; inference is reserved for the held-out evaluation (Section 4.3).

4. WHY THE PROXY SHOULD NOT DECIDE ALONE

Four lines of evidence jointly motivate treating the proxy as a proposal rather than a decision.

4.1 The Gain Is Label-Sensitive

An earlier version used the manifest’s fallback grouping, treating *same-uploader* items as co-members in 26% of examples. On that mixed-label corpus, fusion appeared to beat frozen cosine by $+4.0$ pp on *hard_similar* (949 sources, same protocol). Restricting to true pack co-membership erases the gain (Section 3). The restriction changes the source universe and mined negatives together, so the loss cannot be attributed to the uploader-grouped examples alone; what it shows is that the measured advantage is sensitive to label construction—too sensitive to treat as a reliable utility estimate.

4.2 Regime-Dependent Signal Contributions

Table 1 shows leave-one-signal-out deltas per negative regime. Removing tempo costs -0.96 pp on *hard_similar* but *gains* $+0.68$ pp on *hard_tempo_key*; removing key helps on both hard regimes ($+0.94 / +1.21$ pp). Both replicate on the LAION-CLAP held-out split

(tempo $-1.48 / +0.37$ pp, key $+0.62 / +0.97$ pp); under MS-CLAP, tempo holds ($-6.9 / +0.7$ pp) but key’s *hard_similar* effect reverses (-0.4 pp vs. $+0.9$ pp). Part of this pattern is built in: *easy* and *hard_tempo_key* are constructed from tempo and key, so those signals are near-constant within them by design. Still, a fused score averages over regimes where signals switch sign, so its ranking cannot be uniformly trusted.

4.3 Held-Out Evaluation

The held-out split (121 sources, 54 with queries, 323 examples) was evaluated after development; protocol and split were committed first. Under LAION-CLAP (primary), fusion—cosine on *hard_similar* is $+1.5$ pp with bootstrap 95% CI $[-1.5, +4.7]$ (joint resampling of 54 source clusters)—small and uncertain; *hard_tempo_key* -1.0 pp (CI below zero); overall $+0.4$ pp (CI crossing zero). A sensitivity re-analysis in the deployed MS-CLAP space, retraining with source-weighted BPR (*hard_similar* $\times 2.0$), shows *hard_similar* $+9.2$ pp, CI $[+4.3, +14.3]$ —entirely above zero; *hard_tempo_key* -1.8 pp (CI $[-4.5, +0.6]$); overall $+2.0$ pp, CI $[+0.6, +3.5]$. The 7.7 pp swing on *hard_similar* between spaces evidences representation sensitivity. A dev-set 2×2 factorial (representation $\times$ training recipe, 5-fold CV on train sources) decomposes the fusion—cosine delta: embedding space $+1.9$ pp, source-weighted BPR $+0.5$ pp, interaction $+0.3$ pp—primarily representation-driven.

4.4 Exploratory Pilot

Nine external raters labeled 20 blind A/B pairs (Fleiss’ $\kappa = 0.294$). One pair whose majority chose “neither” was excluded; 9 of the remaining 19 preference pairs reached $\geq 7/9$ consensus. On these 9 pairs, harmonic compatibility—whose weak-label contribution was near zero in the earlier mixed-label 7-signal ablation—matches the human majority in 7/9 cases, while tempo, the strongest *hard_similar* contributor in Table 1, matches in 3/9—a descriptive divergence from the weak-label ranking. The comparison is exploratory—post-hoc, small, and on a different corpus and pair construction—raising the possibility of misalignment with human judgment without confirming it.

5. CONCLUSION

Pocket Producer operationalizes session-conditioned retrieval in a DAW. The fusion is an *operational* cold-start candidate generator whose held-out advantage swings from $+1.5$ pp on *hard_similar* (CI crossing zero, LAION-CLAP) to $+9.2$ pp (CI above zero, MS-CLAP)—representation-sensitive and regime-dependent. It surfaces per-signal evidence; since its ranking cannot be uniformly trusted, every recommendation is previewable and reversible. Evaluating whether preview and undo improve creative outcomes, and personalization from logged feedback, is future work.

- 166 **Acknowledgements.** The author thanks the Audiotool 218
 167 team for the Nexus SDK and the FSLD creators [12]. 219
- 168 **6. REFERENCES** 220
- 169 [1] B.-Y. Chen, J. B. L. Smith, and Y.-H. Yang, “Neural 223
 170 loop combiner: Neural network models for assessing 224
 171 the compatibility of loops,” in *Proceedings of the 21st* 225
 172 *International Society for Music Information Retrieval* 226
 173 *Conference (ISMIR)*, 2020. 227
- 174 [2] S. Lattner, “SampleMatch: Drum sample retrieval by 228
 175 musical context,” in *Proceedings of the 23rd Interna-* 229
 176 *tional Society for Music Information Retrieval Confer-* 230
 177 *ence (ISMIR)*, 2022. 231
- 178 [3] X. Yi and M. Barthet, “A generative framework for 232
 179 composition-aware loop recommendation in music pro- 233
 180 duction: Drum2Bass use case,” in *Proceedings of the* 234
 181 *Sound and Music Computing Conference (SMC)*, 2024.
- 182 [4] S. Roy Chaudhuri, S. P. Madupu, K. T. Guru Sai, 235
 183 and V. Jamwal, “LoopMatcher: Proof-of-concept for 236
 184 AI-assisted music loop search,” in *Proceedings of the* 237
 185 *15th International Conference on Computational In-* 238
 186 *telligence in Music, Sound, Art, and Design (Evo-* 239
 187 *MUSART)*, 2026.
- 188 [5] R. Ciranni, G. Mariani, M. Mancusi, E. Postolache, 240
 189 G. Fabbro, E. Rodolà, and L. Cosmo, “COCOLA: 241
 190 Coherence-oriented contrastive learning of musical au- 242
 191 dio representations,” in *IEEE International Conference* 243
 192 *on Acoustics, Speech and Signal Processing (ICASSP)*, 244
 193 2025.
- 194 [6] J. Nistal, M. Pasini, C. Aouameur, M. Grachten, and 245
 195 S. Lattner, “Diff-A-Riff: Musical accompaniment co- 246
 196 creation via latent diffusion models,” in *Proceedings of* 247
 197 *the 25th International Society for Music Information* 248
 198 *Retrieval Conference (ISMIR)*, 2024.
- 199 [7] A. Riou, A. Gagnere, G. Hadjeres, S. Lattner, and 249
 200 G. Peeters, “Zero-shot musical stem retrieval with 250
 201 joint-embedding predictive architectures,” in *IEEE In-* 251
 202 *ternational Conference on Acoustics, Speech and Sig-* 252
 203 *nal Processing (ICASSP)*, 2025.
- 204 [8] Q. He, G. Xia, and Z. Wang, “TOMI: Transforming 253
 205 and organizing music ideas for multi-track composi- 254
 206 tions with full-song structure,” in *Proceedings of the* 255
 207 *26th International Society for Music Information Re-* 256
 208 *trieval Conference (ISMIR)*, 2025.
- 209 [9] A. C. Elkins, S. Singh, A. Kieback, S. Blankenship, 257
 210 U. P. Amadasun, and A. Chadha, “DAWZY: A new 258
 211 addition to AI powered “human in the loop” music co- 259
 212 creation,” in *NeurIPS Workshop on AI for Music*, 2025.
- 213 [10] B. Elizalde, S. Deshmukh, M. Al Ismail, and H. Wang, 260
 214 “CLAP: Learning audio concepts from natural lan- 261
 215 guage supervision,” in *IEEE International Conference* 262
 216 *on Acoustics, Speech and Signal Processing (ICASSP)*, 263
 217 2023.
- 218 [11] S. Rendle, C. Freudenthaler, Z. Gantner, and 264
 219 L. Schmidt-Thieme, “BPR: Bayesian personalized 265
 220 ranking from implicit feedback,” in *Proceedings of the* 266
 221 *25th Conference on Uncertainty in Artificial Intelli-* 267
 222 *gence (UAI)*, 2009.
- 223 [12] A. Ramires, F. Font, D. Bogdanov, J. B. L. Smith, Y.-H. 268
 224 Yang, J. Ching, B.-Y. Chen, Y.-K. Wu, W.-H. Hsu, and 269
 225 X. Serra, “The freesound loop dataset and annotation 270
 226 tool,” in *Proceedings of the 21st International Society* 271
 227 *for Music Information Retrieval Conference (ISMIR)*, 272
 228 2020.
- 229 [13] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, 273
 230 and S. Dubnov, “Large-scale contrastive language- 274
 231 audio pretraining with feature fusion and keyword- 275
 232 to-caption augmentation,” in *IEEE International Con-* 276
 233 *ference on Acoustics, Speech and Signal Processing* 277
 234 *(ICASSP)*, 2023.